

Crown Melbourne — Kitchen Equipment

Six-Month Work Order Data Review

Pre-meeting briefing and data quality findings

1. Executive Summary

This document covers what I found in the two source files I have been working with, what each column means, the data quality issues I want to raise with the team, and how the demo dashboard uses each piece of information. The numbers in this document have been verified against the raw files and are accurate as of the current extract.

The dataset covers 599 work orders raised across Crown Melbourne kitchens between 28 July 2025 and 4 May 2026 — approximately ten months. Every work order in this extract was assigned to a single vendor, PTL Equipment Services P/L. Total reconciled spend across the period is AUD 303,422.96 across 317 costed work orders.

The two files I am working with are:

- **6_Months_Data.xlsx** — the operational logbook from Maximo. 599 rows, 20 columns. One row per work order. No cost data.
- **6_month_WO_asset_cost.xlsx** — the finance reconciliation. Two sheets ("With Asset" and "Without Asset"), 317 costed work orders, with the dollar amounts. No status data.

The two files are joined on the Work Order number.

2. Source System Context

The column naming convention (WASSIGN, WREVIEW, Owner Group, MELB site code, the seven-digit sequential work order IDs) is consistent with IBM Maximo, which is the enterprise asset management platform Crown uses. The cost file is structured as a manual reconciliation rather than a system export — row 1 of each sheet is a banner heading and the actual column headers sit in row 2. This format suggests the cost file is prepared by hand from a separate finance or accounts-payable system and matched against Maximo work orders. Confirming this with the team would be useful.

3. Main File: 6_Months_Data.xlsx

The file has one sheet containing 599 data rows under 20 column headers. I will walk through each column in order, noting both what it contains and what it does not.

Column 1 — Work Order

The unique identifier for each ticket. 598 of the 599 values are seven-digit integers ranging from 1,860,603 to 1,968,341 — these are sequential, system-generated Maximo IDs.

One row (WO1116) uses a string format with a "WO" prefix and is six characters long. This is the only manual or non-standard ID in the dataset. I would like to ask the team whether this is a test record, a manually created ticket, or part of a different workflow.

Column 2 — Owner Group

This column is intended to identify the team responsible for the work. The values found are inconsistent:

- MAINTMNGRBLD (Building Maintenance Manager team) appears in 113 rows.
- 50121 appears in 1 row — this is the vendor code for PTL, not a valid owner group, which suggests a data-entry error.
- 485 rows (81 percent of the dataset) have no Owner Group recorded at all.

The very high null rate indicates either that the field is not consistently populated during ticket creation, or that the routing logic in Maximo does not assign one automatically. This is worth raising.

Column 3 — Priority

Numeric urgency on a 1 to 6 scale, with 1 being most urgent. The distribution is:

Priority	Count	Interpretation
1	4	Critical — safety or operational emergency
2	355 (59%)	High — equipment down, affects service
3	98	Medium — degraded but workable
4	20	Low — minor issue
5	121	Scheduled / routine
6	1	Lowest

The four Priority 1 (critical) work orders are worth flagging because three of them describe burning-smell or smoke incidents on cooking equipment:

- WO 1965085 (GRADI): "hot box plate has a burning smell - can see smoke"
- WO 1964551 (Room Service): "asian section microwave sparks & smell likes something is burning - HACCP"
- WO 1957805 (PIT9 / Banquets): "New door seal required on Combi Oven"
- WO 1863976 (88 Noodle): "microwave in kitchen has a burning smell inside"

Three of four critical jobs being safety incidents on cooking equipment is a pattern worth raising with the team — it may warrant a safety review or a closer look at the preventive maintenance schedule for ovens and microwaves.

Column 4 — Description

Free-text description of the issue. All 599 rows have a description filled in. Length ranges from 18 to 100 characters, averaging 63 characters. Sample entries:

- "Hot box seal to be replaced"
- "convo oven is error E013000 indicating low water issue with the oven."

- "door seal water leaking from oven - steam leaks out when in use"

This is the most analytically important column in the entire file because there is no structured equipment-type field anywhere else. To produce equipment-level reporting (ovens vs ice machines vs dishwashers, for example), I have had to extract keywords from this free text in the dashboard. That works but it is fragile — variations like "combi", "convotherm" and "convo oven" all refer to the same equipment type but parse differently. I would like to ask the team whether Maximo holds a structured equipment classification that we could pull instead.

Columns 5 and 6 — Location and Location Description

Location is a short code; Location Description is the full hierarchical path. There are 70 distinct location codes in the dataset and every row has both fields filled. Examples:

Location Code	Full Description
HOTKITCH	Mel-Podium-PO1-F&B-BANQBOH-hot kitchen (Production Kitchen)
CONSERVATORY	Mel-Towers Hotel-CT1-F&B-Conservatory
MRHIVE	Mel-Towers Hotel-CTG-F&B-Mr Hive Kitchen & Bar
IDSBOH	Mel-Podium-PO1-F&B-ID's kitchen
MAHOGANYREST	Mel-Towers Hotel-CT1-F&B-Mahogany Restaurant & Kitchen

The naming convention follows a hierarchy: Building → Floor → Category → Specific outlet. So "Mel-Podium-PO1-F&B-BANQBOH-hot kitchen" decodes as Melbourne building, Podium tower, Floor PO1, Food & Beverage, Banquet Back-of-House, hot kitchen.

The five locations generating the most work order volume are IDSBOH (60 work orders), CONSERVATORY (51), HOTKITCH (50), MRHIVE (32) and MAHOGANYREST (23). This column is clean and reliable.

Columns 7 and 8 — Asset and Asset Tag

This is the area where I want the team's attention most. The Asset column is meant to identify the specific piece of equipment a job relates to. Three different formats appear in the column:

Format	Count	Example	Likely meaning
NULL (no asset)	359 (60%)	—	No asset linked when ticket was raised
Numeric ID	188 (31%)	123882, 77424	Maximo system asset ID
Alphanumeric code	52 (9%)	A0311, A0332, A0252	Appears to be vendor asset code (clusters around ovens)

The 60 percent null rate is the single most material data quality finding from this review. The implication is that for the majority of work orders, the maintenance spend cannot be attributed to a specific piece of equipment, which prevents lifecycle costing, replace-vs-repair decisions, and warranty tracking. Looking at the descriptions of unlinked rows, staff are often writing phrases such as "no asset #" or "hot

box (no asset)" directly in the description field, which confirms they know which piece of equipment is involved but the asset is not in the system.

The coexistence of numeric IDs and alphanumeric A-codes (the A0xxx series clusters heavily on ovens) suggests two parallel identifier systems — one from Maximo and one from the vendor. I cannot reliably tell from this data alone whether asset A0311 and Maximo asset 123882 might refer to the same physical oven.

Asset Tag (column 8) is 591 of 599 null. The 8 populated rows contain technical specifications such as "C4eT6:10 ES 3N~ 400V 50/60Hz", which appears to be a Convotherm oven model specification (voltage and phase). This is not an asset tag in the conventional sense; it looks like a freeform notes field that has been repurposed.

DISCUSSION POINT — Asset identification: We have three identifier systems running in parallel (Maximo numeric IDs, vendor A-codes, and free-text Asset Tags). I would like to understand which is the source of truth, and whether they can be unified.

Column 9 — Status

This describes where a ticket sits in its workflow. The six values found are:

Status	Count	Meaning	Avg days from report to status
CLOSE	333 (56%)	Administratively closed and archived	189 days
WASSIGN	108 (18%)	Waiting to be assigned to a technician	40 days
COMPLETE	98 (16%)	Work done but not yet administratively closed	28 days
CANCELLED	39 (7%)	Ticket killed without completion	132 days
WREVIEW	20 (3%)	Waiting for review or approval	57 days
WACTION	1	Waiting for an action	0 days

Several observations are worth raising with the team:

The 189-day average between Reported Date and Status Date for the CLOSE bucket is not the time taken to fix the equipment. Jobs typically pass through COMPLETE first (averaging 28 days from report), then sit in COMPLETE for a long time before being administratively closed. The 189-day figure therefore measures administrative backlog, not maintenance performance. Even so, that is a very long tail.

The WASSIGN queue contains 108 work orders, and within that queue I found tickets that have been sitting unassigned for more than five months. Specific examples:

- WO 1901639 (CROWNELB): "Bratt Pans Pressure Test - Quarterly PM" — reported 23 Nov 2025, still unassigned 5 May 2026 (162 days).
- WO 1901969 (CONFPANTRYE): "PTL to follow up previous WO 1891965 regarding hot water unit" — reported 24 Nov 2025, still unassigned (161 days).

- WO 1921542 (CRYSTALCLUB): kitchen aid mixer noise issue — reported 22 Jan 2026, still unassigned (103 days).

The 39 cancellations include some that were cancelled 200 to 270 days after the original ticket was raised, which looks like cleanup of a stale queue rather than active workflow management.

Column 10 — Reported Date

The date and time the ticket was first raised. The full range in the extract is 28 July 2025 to 4 May 2026. Monthly volumes are steady, with no clear seasonal pattern:

Month	Work Orders
Jul 2025	14 (partial month)
Aug 2025	56
Sep 2025	71
Oct 2025	78
Nov 2025	61
Dec 2025	50
Jan 2026	78
Feb 2026	74
Mar 2026	58
Apr 2026	55
May 2026	4 (partial month)

Steady-state volume is roughly 60 to 78 tickets per month, averaging about two new tickets per day.

Columns 11 and 12 — Contract Manager and Owner

Both columns are almost entirely empty:

- Contract Manager: 598 of 599 rows are null. One row contains the value 434948.
- Owner: 599 of 599 rows are null.

Either these fields are not used in Crown's Maximo workflow, or they were excluded from the export. I would like to confirm which, because accountability tracking without an owner field is difficult.

Columns 13 and 14 — Vendor and Vendor Name

Every row in the dataset shows vendor code 50121 (PTL Equipment Services P/L). There is no other vendor in this extract. The vendor name contains non-breaking space characters between words (a Maximo export quirk that is visually invisible but can break exact-match filters in some tools). The dashboard handles this correctly.

It would be useful to confirm whether PTL is genuinely the only kitchen equipment vendor at Crown Melbourne, or whether work for other vendors has been filtered out of this extract.

Column 15 — Affected User

The Crown staff member who reported the issue. 598 of 599 rows are null; only one row has a name recorded (Mark Gurung, on the anomalous WO1116). The field appears to be either unused or not captured during ticket creation.

Column 16 — Timer Status

SLA timer field. 596 of 599 null, with three rows marked COMPLETE. Effectively unused.

Column 17 — Status Date

The date the status field last changed. Combined with Reported Date, this is what allows resolution-time analysis. The values used in the status duration table above were calculated from this column.

Column 18 — Work Type

The type of maintenance work performed:

Type	Count	Meaning
CM	478 (80%)	Corrective Maintenance — reactive, fix after breakage
PM	116 (19%)	Preventive Maintenance — scheduled servicing
PROJ	3	Project work — replacement or installation
INSP	1	Inspection
(null)	1	Missing

The 80 percent corrective / 19 percent preventive split is heavy on the reactive side. Industry benchmarks for hospitality kitchens typically sit closer to 60 percent preventive / 40 percent corrective. The current ratio suggests the PM programme may be under-coverage relative to the equipment base. This usually correlates with higher overall cost and more disruption during peak service hours.

DISCUSSION POINT — Maintenance mix: 80 percent of tickets are reactive corrective maintenance. Is the preventive maintenance programme adequately resourced and scoped to cover the kitchen equipment base?

Column 19 — Site

All 599 rows show site MELB (Crown Melbourne). The extract is single-site.

Column 20 — Qualifications Required

All 599 rows show N. Either no kitchen equipment work requires special qualifications, or the field is not actively used. Worth a quick clarification with the team.

4. Cost File: 6_month_WO_asset_cost_.xlsx

This file contains two sheets. Both sheets share a non-standard layout where row 1 is a banner title ("Crown Work Orders – With Asset | Total Price Reconciliation" or the equivalent for the second sheet) and row 2 holds the actual column headers. Data begins in row 3. I had to adjust the dashboard's parsing logic to handle this — a standard import that assumes row 1 is the header would corrupt the data.

4.1 Sheet "With Asset" — 132 rows, AUD 119,155.34

These are costed work orders where the equipment was properly tagged in Maximo at the time the work was done. The sheet has five active columns: Work Order, Description, Location Code, Asset, and Total Price (inc GST). The price is in Australian dollars, GST-inclusive.

4.2 Sheet "Without Asset" — 185 rows, AUD 184,267.62

These are costed work orders where the equipment was not properly tagged. The sheet is wider (nine columns) but most are unused:

- Work Order, Description, Location Code — same as the With Asset sheet.
- **Asset** — contains the literal text "*Missing*" in approximately 155 rows, the text "*PM*" in roughly 25 rows (preventive maintenance batches that cover multiple assets in a single invoice), and occasionally a numeric ID where the asset was known after the fact.
- **No. of Assets and Avg cost** — both columns are entirely empty in the current dataset. They appear to have been intended as calculation columns for batch jobs (total price divided by number of assets equals average cost per asset) but no values have been populated. I would like to ask the team whether these columns should be calculated or whether the design was abandoned.
- **Total Price (inc GST)** — the actual cost figure for these rows.

4.3 Cost totals

Verified against the raw file twice (once with Python, once with the same JavaScript library the dashboard uses):

Bucket	Work Orders	Total (AUD inc GST)
With Asset	132	\$119,155.34
Without Asset	185	\$184,267.62
GRAND TOTAL	317	\$303,422.96

A material observation: 61 percent of the total spend (AUD 184k of AUD 303k) is associated with work orders where the asset was not properly captured, meaning that spend cannot be tied to specific equipment for lifecycle analysis.

5. How the Two Files Link Together

The two files join on the Work Order number. The actual overlap is as follows:

Set	Count	Interpretation
Work orders in main file	599	Operational logbook entries
Work orders in cost file (combined sheets)	317	Costed and reconciled
Present in BOTH files	267	Full picture — description, status, location AND cost
In main file only	332	Open, cancelled, or just-completed — not yet invoiced
In cost file only	50	Invoiced this period but raised before 28 Jul 2025

6. How the Demo Dashboard Uses Each Column

Below is the column-to-visualisation mapping I implemented. The dashboard reads both Excel files in the browser, joins them on Work Order number, and then computes each KPI and chart on the fly.

Dashboard element	Source columns	Calculation method
Total Work Orders KPI	Work Order	Row count
Total Spend KPI	Total Price (cost file)	Sum across both sheets
CM vs PM ratio	Work Type	Count by category
Completion Rate	Status	(CLOSE + COMPLETE) / total
Cancellation Rate	Status	CANCELLED / total
Avg Resolution Time	Reported Date, Status Date	Date difference, COMPLETE/CLOSE only
Asset Linkage Rate	Asset	Non-null count / total
Monthly Volume Chart	Reported Date	Group by year-month
Priority Distribution	Priority	Count by 1-6
Status Pie Chart	Status	Count by status value
Top Locations Table	Location + Total Price	Count and sum per location
Equipment Keywords Chart	Description	Keyword regex match (oven, ice, door, kettle, etc.)
Cost Distribution	Total Price	Bucket: <\$200, \$200-500, \$500-1k, \$1k-2k, \$2k+
Top Assets by Cost	Asset + Total Price	Group by asset, sum cost
With/Without Asset Split	Sheet name (cost file)	Sum per sheet

Dashboard element	Source columns	Calculation method
Costliest WO Table	All cost rows	Sort by Total Price descending
Location Filter (top bar)	Location	Drives all charts when changed

Columns that the dashboard does not currently use, and why: Owner Group, Contract Manager, Owner, Affected User, Timer Status, Qualifications Required, Site, and Vendor. These are either almost entirely null in the dataset or hold a single repeating value across all 599 rows, so they produce no useful chart. They have been retained in the data layer in case future versions of the dashboard expose them (for example, an owner-accountability view once Owner Group is properly populated upstream).

7. From Demo to Production

The current dashboard is a demo. It reads Excel files dropped into the browser, parses them client-side in JavaScript, and computes everything in memory. This is appropriate for proving the analytical model and getting stakeholder sign-off on the KPIs, but it is not how this should run in production.

The production path is straightforward and the visual layer of the dashboard barely changes:

1. **Ingestion** — a scheduled job (nightly or hourly) pulls work order data from the Maximo API or database and reconciled cost data from the finance or accounts-payable system. This can be done via an ETL platform (Azure Data Factory, Fivetran, Airbyte) or a simple Python script on a cron schedule.
2. **Database** — the data lands in a relational store such as PostgreSQL, SQL Server, Snowflake, or BigQuery. The schema would consist of tables such as work_orders, work_order_costs, assets, and locations, with proper foreign keys so the Work Order to cost join becomes a SQL join rather than a JavaScript loop. The data volumes involved (around 1,200 work orders per year for a single site, perhaps 10,000 to 15,000 across all Crown sites and a few years of history) are modest and any of these databases would handle them comfortably.
3. **API layer** — the dashboard calls a server endpoint such as /api/work-orders?site=MELB&from=2025-07-01 rather than parsing a file. KPI calculations live server-side, so every user sees identical numbers and the dashboard loads instantly.
4. **Same dashboard UI** — the visual layer (charts, KPIs, layout) is essentially unchanged. The only code that changes is the data-loading function: XLSX.read(file) is replaced with fetch('/api/work-orders'). Once stakeholders sign off on the KPIs the demo is showing, the engineering work to wire this in is mechanical.

The demo, however, exposes three things that production needs to address at the source, not at the dashboard layer:

- Asset linkage discipline in Maximo — moving the 60 percent null rate down requires a process change at ticket creation, not a database fix.
- A reconciliation process between Maximo work orders and the finance system, with a defined and reliable join key. The current cost file appears to be a manual reconciliation.
- A structured equipment type field, or a classification step in the ingestion pipeline, so that equipment-level reporting does not rely on free-text keyword parsing.

8. Discussion Points for the Meeting

Pulling the threads together, these are the items I would like to raise with the team:

1. Confirm that the source system is IBM Maximo and that the cost file is a manual reconciliation against the finance / accounts-payable system.
2. Asset linkage: 60 percent of work orders have no Asset ID. This prevents lifecycle costing, replace-vs-repair decisions, and warranty tracking. What is the path to improving this at the ticket creation step?
3. Three coexisting asset identifier systems (Maximo numeric IDs, vendor A-codes such as A0311, and free-text Asset Tags). Which is the source of truth and can they be unified?
4. Maintenance mix is 80 percent corrective / 19 percent preventive. Industry benchmark is closer to 60/40 in favour of preventive. Is the PM programme adequately resourced?
5. The CLOSE status averages 189 days from Reported Date — driven by administrative backlog between work completion and ticket closure, not by maintenance performance. Is this a known issue?
6. 108 work orders are sitting in WASSIGN, some for more than five months including PM jobs. Is there a triage process for the unassigned queue?
7. 39 cancellations, several cancelled 200+ days after raising. Are these duplicates, work-not-required, or budget cuts?
8. Owner Group is missing on 81 percent of work orders. Should this be made mandatory at creation?
9. Three of the four Priority 1 (critical) tickets are burning-smell or smoke incidents on cooking equipment. Worth flagging for safety review.
10. Contract Manager, Owner, Affected User, Timer Status, and Qualifications Required are essentially empty. Are these fields not part of the workflow, or have they been excluded from the export?
11. The cost file's "No. of Assets" and "Avg cost" columns are designed but never populated. Should these be calculated for batch PM jobs covering multiple assets?
12. WO1116 is the only non-numeric work order ID in the dataset. Is this a test record, a manual entry, or part of a different workflow?

End of briefing